SHREE VAISHNAVI M G

Firmware Engineer

+91 8105171167 | vaishnavimudligiri@gmail.com | linkedin.com/in/shree-vaishnavi-m-g | Bengaluru, India

SUMMARY

Embedded Firmware Engineer with hands-on experience developing ESP32-based products in Embedded C and ESP-IDF, covering display, payment, multimedia, storage, communication, audio and wireless update subsystems. Experienced in FreeRTOS, LVGL, SPI/I2C/UART/I2S interfacing, BLE OTA, and hardware bring-up, debugging and validation on physical prototypes, extending into schematic and PCB design.

EXPERIENCE

IoT Full Stack Junior Consultant

Vittam Solutions Pvt. Ltd., Bengaluru 2025 – Present

Smart Payment Kiosk — embedded firmware and hardware development (ESP32 / ESP32-S3)

- Developed firmware for an ESP32-based smart payment kiosk integrating digital advertising, QR payments, content management, audio and wireless updates.
- Built the 320x480 TFT display and multimedia system for continuous promotional image and video playback using SD-card content, LVGL and ESP-IDF.
- Integrated USB and Wi-Fi communication, Android-based payment interaction, QR status display and audio payment confirmation.
- Implemented BLE OTA firmware updates and content-management functionality for remotely deployed devices.
- Performed hands-on hardware bring-up and firmware debugging across display, memory, peripherals and communication; currently extending the system into complete schematic and PCB design through hardware-block analysis and component selection.

Technologies: ESP32, ESP32-S3, ESP-IDF, FreeRTOS, Embedded C, LVGL, ESP GMF, SPI, USB, Wi-Fi, BLE OTA, SD card, JPEG/MJPEG, DMA, PSRAM, SPIFFS

Engineering Intern

Bharat Heavy Electricals Limited (BHEL) 6 weeks

- Industrial exposure to electronics, electrical systems, instrumentation and control systems.

PROJECTS

Custom USB-C Embedded Development Board

Schematic design, firmware

- Designed a custom USB-C powered embedded board integrating an MCU, power regulation, protection and peripheral interfaces.
- Developed firmware for SPI, I2C, UART and GPIO peripherals and performed board-level hardware bring-up and testing.
- Designed the schematic covering power distribution, decoupling, connectors and programming/debug interfaces.

10-bit SAR ADC

90 nm CMOS, Cadence Virtuoso

- Designed a 10-bit SAR ADC in 90 nm CMOS for high-speed, low-power operation at a 1.5 V supply, achieving approximately 900 kS/s sampling rate at 31.18 uW power consumption.

Electromagnetic Robotic Arm (ERA)

Arduino, servos, electromagnet

- Built an Arduino-controlled robotic arm with servos and an electromagnet for automated separation of ferromagnetic material, using PWM control and relay switching for precise pick-and-place operation.

EDUCATION

B.E., Electronics and Communication Engineering
Dayananda Sagar College of Engineering (VTU), Bengaluru
2026 | CGPA 7.8/10

Pre-University (PCMB)
97.3% | 2022

CBSE Class X
92.4% | 2020

SKILLS

Embedded Firmware
Embedded C, C++, ESP-IDF, FreeRTOS, LVGL, ESP GMF, firmware development and debugging

Platforms
ESP32, ESP32-S3, Arduino, Intel FPGA

Interfaces and Peripherals
SPI, I2C, UART, USB, Wi-Fi, BLE, I2S, SDMMC, GPIO, PWM, DMA

Display and Multimedia
TFT LCD, JPEG, MJPEG, RGB565, video and image playback, audio playback

Memory and Storage
SD card, PSRAM, SPIFFS, buffering, memory management

Hardware
Schematic design, PCB design, component selection, power regulation, board bring-up, datasheet analysis

Tools
Git, VS Code, Python, Cadence Virtuoso, Quartus Prime, ModelSim, NI Multisim

CERTIFICATIONS AND AWARDS

- 1st Place, Electromania 2024 — RTL design and problem solving using ModelSim.
- NPTEL — System Design Through Verilog, IIT Guwahati.
- MATLAB Onramp — MathWorks.